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Amaranthus palmeri S. Watson reproduction system: Implications for distribution and management strategies

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Abstract

Amaranthus palmeri S. Watson is one of the most troublesome weed species worldwide, and is considered one of the most devastating weeds in agricultural fields in the United States. Recently, *A. palmeri* populations have spread beyond their native range, into the Mediterranean region, and have been reported in several European countries. Resistance to 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) and acetolactate synthase (ALS) inhibitors have been found in European populations, exacerbating the management problems with this species in the Mediterranean region. While the biology, ecology, herbicide resistance mechanisms, and agricultural significance of *A. palmeri* have been studied, there remains a need to discuss how *A. palmeri*'s unique reproductive traits influence its ability to adapt to various environments, especially as an invasive species spreading into new habitats. We review how the genetic and genomic characteristics of *A. palmeri* interact with aspects of its reproductive biology, including its breeding system, and alter its ability to hybridize and develop herbicide resistance. Finally, we discuss the breeding system of *A. palmeri* in the context of weed management, and explore different approaches such as irradiated pollens, genetic manipulations, and RNA interference (RNAi).

KEYWORDS

dioecious, herbicide resistance, hybridization, invasive plant species, weed control

1 | INTRODUCTION

Amaranthus palmeri S. Watson (Palmer amaranth) is one of the most troublesome weed species worldwide that competes in both agricultural (e.g., field crops, vegetable crops, orchards) and non-agricultural (e.g., field margins, roadsides, railways) habitats (Bagavathiannan & Norsworthy, 2016; Bravo et al., 2017; Massinga et al. 2001). It is native to the southwestern United States and northern Mexico. However, it has spread rapidly beyond its original range across North and South America, Europe, East Asia, and the wider Mediterranean

region (Cao et al. 2022; Dominguez-Valenzuela et al., 2017; Gaines et al., 2021; Manicardi et al., 2025; Milani et al., 2021; Torra et al., 2020). Generally, *A. palmeri* occurrence and spread relate to its close association and dispersal within agricultural systems, broad environmental adaptability, and stress tolerance. This species has several biological attributes that explain its weediness; rapid growth and high seed reproduction rates, tendency to evolve resistance to herbicides, and the ability to reach high abundances in a short time (Heap, 2023; Horak & Loughin, 2000; Norsworthy et al., 2014). Additionally, the dioecious nature of *A. palmeri* makes it an obligatory outcrossed

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species, which facilitates the spread of adaptive traits, including herbicide resistance genes (Sosnoskie et al., 2012).

In the US, *A. palmeri* is among the most common weed species and is ranked as the most troublesome weed in agricultural fields (van Wychen, 2017). Weed species belonging the *Amaranthus* genus have the Kranz anatomy and C₄ photosynthetic pathway (Wang et al., 1992), of which carbon fixation is more efficient compared to C₃ and crassulacean acid metabolism (CAM) plants. Interestingly, *A. palmeri* displays faster growth rate than other closely related species, and generally accumulates greater amount of biomass (Horak & Loughin, 2000; Sellers et al., 2003). In broadleaf crops, such as soybeans and peanuts, it can reduce yields by more than 37% and 78%, respectively (Basinger et al., 2019; Hare et al., 2019). *Amaranthus palmeri* also results in high indices of yield losses when growing in competition with maize (*Zea mays* L.) and grain sorghum (*Sorghum bicolor* (L.) Moench.) (Massinga et al., 2001; Moore et al., 2004).

Amaranthus palmeri's damage is not restricted to a specific geography or growing seasons, as weed seeds may be transferred via agricultural machinery and import-export trades between countries (Buddenhagen et al., 2021). Generally, bulk import of grains for various purposes is not always under strict agriculture quarantine regulations, resulting in a continuous seed-mediated gene flow of *A. palmeri* populations. For instance, in Japan, a glyphosate-resistant *A. palmeri* population was found near grain-import ports (Shimono et al., 2020). In Spain, a glyphosate-resistant *A. palmeri* population was collected from a roadside near a commercial port in Tarragona (Manicardi et al., 2025).

Amaranthus palmeri is a highly adaptable species with remarkable phenotypic plasticity and genetic diversity (Bravo et al., 2017; Küpper, Manmathan, et al., 2018). This adaptability can be attributed to important traits concerning its reproduction, such as (1) high seed producer, which explains why it rapidly reaches large population sizes, (2) dioecious mating system, and, therefore, and obligate outcrossing species, and (3) it is wind pollinated, with pollen spreading over long distances (Sauer, 1955; Sosnoskie et al., 2012; Webster & Grey, 2015). Nowhere do we see *A. palmeri*'s adaptability on display more than in cases of herbicide resistance evolution. To date, populations of *A. palmeri* have evolved resistance to nine herbicide sites of action (SOA) and, due to the sexual traits listed above, these herbicide resistance mechanisms spread to susceptible populations as well as stack rapidly in multiple resistant individuals (Gaines et al., 2020).

Amaranthus palmeri's range is expected to continue expanding largely due to its broad climatic and stress tolerance. As an example, studies on the effect of water stress on *A. palmeri*'s seed production showed that drought affected plant biomass, but plants were able to produce viable seeds that, in some cases, possessed inherited drought tolerance (Matzrafi et al., 2020). Additionally, Chahal et al. (2018) showed that *A. palmeri* can survive under conditions of continued water stress and produce a large number of seeds, with no effect on seed viability.

Although the biology, ecology, resistance mechanisms, and agricultural significance of *A. palmeri* have been studied and reviewed in several manuscripts, there remains a need to unify *A. palmeri*'s unique

reproductive traits with its ability to adapt to various environments and stresses including the invasion of new habitats. The objective of this review is to fill this gap, specifically by discussing how its reproductive system affects evolution of weediness traits and invasion potential.

2 | POPULATION GENETICS AND GENOMICS

2.1 | Sex chromosomes in *A. palmeri*

Data regarding the emergence of sex chromosomes as an evolutionary process in dioecious species is limited, especially for the *Amaranthus* genus, where both dioecious and monoecious species exist. It is likely that dioecy evolved independently from hermaphroditic ancestors in different plant families and genera (Ming et al., 2007). Using whole-genome approaches to assemble male-specific sequences, researchers identified a region of approximately 1.3–2 Mb associated with sex (Montgomery, Giacomini, Weigel, & Tranel, 2021; Neves et al., 2020), and have proposed a XY chromosomal model for sex-determination system in *A. palmeri*. In *A. palmeri* and *A. tuberculatus*, respectively, 345 and 2754 male-specific sequences, were identified designing sex-specific markers for both species with high degree of accuracy (Montgomery et al., 2019). The recent genome sequencing and explorations of male and female transcriptomes and genomes have provided male-specific Y molecular markers, insights on sex determination, and how dioecy evolved independently in *A. tuberculatus* and *A. palmeri*. Subsequently, a genome assembly of *A. palmeri* revealed that the male-specific Y region is approximately 1.3 Mbp long with 121 predicted genes, whereas in *A. tuberculatus* it is 4.6 Mbp and contains 147 genes (Montgomery, Giacomini, Weigel, & Tranel, 2021). Male *A. palmeri* and *A. tuberculatus* male-specific Y regions appeared to not be syntenic, suggesting that dioecy evolved independently in both species, leading to the hypothesis that dioecy has evolved multiple times in this genus (Montgomery, Giacomini, Weigel, & Tranel, 2021). The implications of these findings are not restricted to basic science and the evolution of sex-determination system. We will provide some suggestions later in this review about how the genetics of sex determination could be an 'Achilles heel' to target for RNAi gene silencing.

2.2 | Hybridization with close *Amaranthus* species, molecular methods to explore differences among species

Identification of *Amaranthus* species based on morphology can sometimes be difficult due to similarities among them, especially in young growth stages and seeds (Matzrafi et al., 2025). However, it is highly important since different species have distinct mating systems and respond differently to management practices. Interspecific crosses have been demonstrated to occur under experimental conditions

between several *Amaranthus* species, and hybridization deserves special attention as hybrids can result in the transfer of alleles associated with adaptation (e.g., herbicide resistance) from one species to another, making weed management more difficult. The rate of interspecific hybridization between *A. Palmeri* and other *Amaranthus* species has been investigated in many studies. Gaines et al. (2012) reported that, in field experiments, hybridization rates from 0.01% to 0.4% occurred between a glyphosate-resistant male *A. palmeri* and *A. spinosus*, less than 0.2% with *A. tuberculatus*, and less than 0.01% with *A. hybridus*. In greenhouse conditions, *A. palmeri* × *A. spinosus* hybridization rate was up to 1.4%. Increased 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) copy number, a mechanisms of glyphosate resistance, was found to be inherited in some of the hybrid offspring (detected through q-RT-PCR). Interestingly, hybrids between *A. palmeri* and *A. hybridus* show male and female plants, whereas hybrids of *A. palmeri* and *A. spinosus* show male and monoecious inflorescences (Gaines et al., 2012). *Amaranthus palmeri* × *A. powellii* and *A. palmeri* × *A. retroflexus* hybrids were not identified in the same study, and to the best of our knowledge, neither they have been reported elsewhere.

Several PCR-based molecular markers have been developed to confirm the occurrence of interspecific hybridization in *Amaranthus* species. Internal transcribed spacer 1 (ITS1) and ITS2 sequences and restriction enzymes can successfully distinguish eight *Amaranthus* species (*A. albus*, *A. arenicola*, *A. blitoides*, *A. hybridus*, *A. palmeri*, *A. powellii*, *A. spinosus*, and *A. retroflexus*), with *A. rudis* and *A. tuberculatus* displaying unique digestion profiles from the others, but not distinct from each other (Wetzel et al., 1999). Similarly, a Kompetitive Allele Specific PCR (KASP) assay has been used to detect hybrids between *A. palmeri* and *A. tuberculatus* and gene flow of mesotrione resistance (Oliveira et al., 2018). The acetolactate synthase (ALS) gene is known to have species-specific polymorphisms and have been used for species determination and hybrids detection. For example, position 678 in this gene has a thymine in *A. palmeri* and a cytosine in *A. tuberculatus* (Tranel, & Wright, 2002). This has been used to differentiate both species as well as their hybrids through a restriction enzyme (MfeI) at that locus (Franssen et al., 2001). Also, a KASP assay based on the same ALS single nucleotide polymorphism (SNP) has been used for species determination in a population of dioecious *Amaranthus* species introduced in Brazil (Küpper et al., 2017). Recently, an *A. palmeri*-specific SNP was identified that can be used to distinguish a seed of *A. palmeri* in up to 200 seeds of other mixed *Amaranthus* species (Brusa et al., 2021).

Cytogenetic tools have also been explored for *Amaranthus* identification. Flow cytometry identified a mean 2C genome size values of 1.04 and 1.34 pg, respectively, for *A. hybridus* and *A. tuberculatus* (Jeschke et al., 2003). *Amaranthus palmeri* genome size was estimated to be the smallest (0.96 pg) when compared to five other *Amaranthus* species (2C values ranging from 1.04 to 1.23 pg) (Rayburn et al., 2005).

With the reduction of costs of sequencing techniques, we expect that more sequence data becomes available for most *Amaranthus* species. Species-specific markers can then be developed to determine *Amaranthus* species and their hybrids, particularly in young plants or

seeds, as demonstrated in the case of *A. palmeri* (Montgomery et al., 2019). In most of the cases described here, hybrids have been studied using bi-parental crosses with herbicide-resistant and -susceptible progenitors, since gene flow of resistance is relatively easy to assess phenotypically. However, the ability to detect hybrids occurring naturally, where more than one species is present, might be critical for designing effective control practices. Purely relying on herbicide resistance as a marker of putative hybrids has some limitations as it requires accurate application and phenotyping. Other limitations of these detection methods are that they depend on the mechanism of resistance and the dominance of that trait in heterozygous individuals (Gaines et al., 2012; Montgomery, Giacomini, & Tranel, 2021). It should also be noted that facultative apomixis can occur in *A. palmeri*, where tissue from the ovule is transformed into an embryo, avoiding the processes of meiosis and fertilization (Ribeiro et al., 2014), further complicating hybrid studies when studying gene flow from a different species into a female *A. palmeri* individual. Therefore, for interspecific hybridization diagnostics, we encourage utilizing multiple techniques for more robust conclusions.

3 | THE BREEDING SYSTEM OF *A. PALMERI*

Flowering plants have evolved a diversity of sexual systems that can range from almost completely self-pollinated to exclusively outcrossing (i.e., dioecy) (Barrett & Harder, 2017; Richards, 1997). The evolution of any specific sexual strategy is accompanied by genetic and/or demographic trade-offs that shape a species ability to adapt to new environments. Although dioecy maximizes outcrossing and thereby reduces the likelihood of inbreeding depression, it may have several disadvantages. For example, fertilization occasionally fails due to mismatch in time or space between male and female plants due to differences in timing of anthesis (Mesgaran et al., 2021). Moreover, even when successful, half of the population (i.e., male plants) produces no seeds. The fecundity of a dioecious population is sensitive to the changes in the sex ratio (the proportion of male/female plants) and, theoretically, the population is likely to decline if it becomes male or female biased (Liu et al., 2021). Thus, the smaller the degree of temporal overlapping between male and female flowering, the lower the chance of successful fertilization and hence seed production.

As a dioecious species, *A. palmeri* individuals reproduce unisexually with male and female flowers on different plants. Studies related to the high plasticity of *A. palmeri* point to its dioecy as a way to explain its large genetic variation and phenotypic plasticity (Küpper, Manmathan, et al., 2018; Sosnoskie et al., 2012). Zhou and Chen (2010) suggested that both pollen-mediated gene flow and seed dispersal may affect inbreeding and genetic structure of a population. Thus, to estimate the consequences for gene movement and spatial genetic structure, the magnitude and scale of seed and pollen dispersal needs to be measured for a specific population. Apart from pollen-mediated gene flow and seed dispersal, the degree of genetic variability is highly influenced by the size of local population and female: male sex ratio. As specified above, anthesis mismatch and

different sex ratio in a population may have an extensive effect on the reproductive capabilities of *A. palmeri*, as observed in a study with 50 individual plants (Mesgaran et al., 2021).

4 | DIOECY AND HERBICIDE RESISTANCE EVOLUTION

4.1 | Inheritance of herbicide resistance traits in *A. palmeri*

To date, *A. palmeri* has evolved resistance to nine herbicide SOAs: microtubule, ALS (Horak & Peterson, 1995; Nakka, Thompson, Peterson, & Jugulam, 2017), EPSPS (Culpepper et al., 2006), glutamine synthetase (GS) (Priess et al., 2022), protoporphyrinogen oxidase (PPO) (Salas et al., 2016), photosystem II (PSII) (Nakka, Godar, Thompson, et al., 2017), 4-hydroxyphenylpyruvate dioxygenase (HPPD) (Nakka, Godar, Wani, et al., 2017), and very long-chain fatty acid (VLCFA) (Brabham et al., 2019), microtubule inhibitors (Gossett et al., 1992) as well as auxin mimics (Shyam et al., 2022). A variety of resistance mechanisms have been identified in *A. palmeri* which may vary based on a population's specific evolutionary history, or the selection pressures that were applied (Table 1). In *A. palmeri*, target-site resistance (TSR) mechanisms include SNPs, codon deletions, and copy number variation of genes encoding herbicide target enzymes, and are commonly found for ALS, PPO, EPSPS and PSII inhibitors (Table 1). The major non-target site resistance (NTSR) mechanism identified in *A. palmeri* is increased herbicide detoxification, which was found to confer resistance to inhibitors of ALS, HPPD, PSII, VLCFA, PPO, and auxin mimic (Table 1). Understanding the resistance mechanisms for various modes of action is critical for unravelling aspects of weed evolution, and how these mechanisms spread. For instance, while mutations in nuclear genome-encoded genes can segregate following a single gene model, NTSR mechanisms can be quantitative (Délye, 2013). Furthermore, many NTSR mechanisms have not been fully characterized and, in most cases, their inheritance patterns remain unknown.

A crucial factor that determines resistance inheritance is the location of the resistance gene, whether it is on the nuclear, chloroplastic, or mitochondrial genome. Nuclear encoded genes are transferred to the progeny after recombination, while the chloroplastic and mitochondrial genomes are typically inherited maternally. For instance, in a PSII inhibitor-resistant *A. palmeri* population, atrazine is conjugated with glutathione by a single nuclear gene (Nakka, Godar, Thompson, et al., 2017; Shyam et al., 2021). However, resistance to PSII inhibitors in other weed species is often caused by mutations in the chloroplastic gene *psbA*, including in some closely related *Amaranthus*, such as *A. retroflexus*, *A. hybridus*, and *A. powellii* (Davis et al., 2019; Dumont et al., 2016; Hirschberg & McIntosh, 1983; Park & Mallory-Smith, 2006). The maternal inheritance of chloroplastic alleles can lead to the rapid fixation of mutations.

Mutation in the ALS (A122, P197, T574, and S653) and PPO2 (A128, ΔG210, V361, and G399) have been found in resistant

A. palmeri populations (Giacomini et al., 2017; Küpper et al., 2017; Larran et al., 2017; Montgomery, Giacomini, & Tranel, 2021; Nakka, Thompson, Peterson, & Jugulam, 2017; Nie et al., 2023; Rangani et al., 2019; Salas et al., 2016; Singh et al., 2019). The ALS and PPO2 are examples of enzymes that function in plastids and mitochondria, but are nuclear-encoded genes. The degree of dominance of ALS and PPO2 resistance alleles may differ among plant species and/or alleles (Foes et al., 1999; Hart et al., 1993; Sebastian et al., 1989; Wright & Penner, 1998), but these were demonstrated to be at least partially dominant (Patzoldt et al., 2006; Yu & Powles, 2010; Brusamarello et al., 2020), and can be selected for in heterozygous individuals (Giacomini et al., 2017; Küpper et al., 2017).

Enhanced herbicide detoxification, a type of NTSR, has been hypothesized to often be a quantitative trait (Délye, 2013). However, in an atrazine-resistant population the herbicide was found to be conjugated by a glutathione S-transferase enzyme, and further studies revealed that this trait followed a single nuclear gene segregation model (Nakka, Godar, Thompson, et al., 2017; Shyam et al., 2021). Inheritance studies have demonstrated that metabolism of chlorsulfuron and mesotrione are conferred by single and multiple nuclear genes, respectively (Shyam et al., 2021). A recently discovered dicamba-resistant *A. tuberculatus* trait is potentially polygenic, and although the resistance was predominantly inherited as a nuclear trait, minor maternal effects were not completely ruled out (Bobadilla et al., 2022). Given the complex and potentially polygenic inheritance of NTSR, predicting its inheritance is difficult and there are no rules-of-thumb that apply across all resistance cases.

Amaranthus palmeri stands out from many other weeds by the sheer amount of multiple resistance cases reported (Heap, 2023). It has been demonstrated that target-site resistance mechanisms can spread quickly within and across *A. palmeri* populations, due in large to copious amounts of wind-blown pollen and obligatory outcrossing in this dioecious species (Jhala et al., 2021; Sosnoskie et al., 2012; Ward et al., 2013). Although these traits are typically single genes, they may stack in one individual through time leading to multiple resistance.

4.2 | The unique inheritance of glyphosate resistance in *A. palmeri*

Glyphosate resistance was first identified in *A. palmeri* in 2005 (Culpepper et al., 2006) and quickly spread throughout the Southern US and across other regions (Heap, 2023). In 2010, a novel and unique TSR mechanism was discovered: EPSPS gene amplification, also known as copy number variation (CNV). In the case of EPSPS-CNV, resistant plants can have 30 to >150 copies of the gene. More copies of the EPSPS result in greater transcript and protein abundance such that the recommended dose of glyphosate no longer effectively inhibits all EPSPS in the cells (Gaines et al., 2010; Gaines et al., 2011). Despite glyphosate-resistant individuals displaying a massive copy number variation and EPSPS expression, no fitness penalty has been detected in greenhouse experiments (Giacomini et al., 2014; Vila-Aiub

TABLE 1 Reported cases of resistance in *A. palmeri* including the mechanism of resistance.

Site of action	Mechanism of resistance ^a	Country	Reference ^b
ALS-inhibitors (HRAC 2)	A122T/S	Argentina, United States	Larran et al. (2017), Palmieri et al. (2022), Singh et al. (2019)
	P197L/A	Spain, United States	Kohrt et al. (2017), Küpper et al. (2017), Singh et al. (2019), Manicardi et al. (2023)
	A205V	Argentina	Palmieri et al. (2022)
	N376E	Italy, Spain	Manicardi et al. (2023)
	W574L	Argentina, Brazil, Israel, Italy, Spain, United States	Abu-Nasar et al. (2025), Gonçalves Netto et al. (2016), Küpper et al. (2017), Larran et al. (2017), Manicardi et al. (2023), Milani et al. (2021), Singh et al. (2019), Torra et al. (2020)
	S653N	Argentina, Brazil, South Africa, Spain, United States	Berger et al. (2016), Gonçalves Netto et al. (2016), Larran et al. (2017), Manicardi et al. (2023), Reinhardt et al. (2022), Singh et al. (2019), Torra et al. (2020)
	Enhanced detoxification	United States	Nakka, Thompson, Peterson, and Jugulam (2017)
	ND	Greece	Kanatas et al. (2021)
Microtubule inhibitors (HRAC 3)	ND	United States	Gossett et al. (1992)
Synthetic auxins (HRAC 4)	Enhanced detoxification	United States	Shyam et al. (2022)
	ND	United States	Foster and Steckel (2022)
PSII-inhibitors (HRAC 5)	Enhanced detoxification	United States	Nakka, Godar, Thompson, et al. (2017)
EPSPS-inhibitor (HRAC 9)	EPSPS amplification	Argentina, Brazil, China, Japan, South Africa, Spain, United States, Uruguay	Cao et al. (2022), Gaines et al. (2010), Gaines et al. (2020), Koo et al. (2018), Küpper et al. (2017), Manicardi et al. (2025), Reinhardt et al. (2022), Shimono et al. (2020)
	EPSPS overexpression (without amplification)	Argentina	Kaundun et al. (2019)
	P106S	Argentina, Mexico	Dominguez-Valenzuela et al. (2017), Kaundun et al. (2019)
	Reduced absorption and/or translocation	Argentina, Mexico	Dominguez-Valenzuela et al. (2017), Kaundun et al. (2019)
	ND	Turkey, United States	Mennan et al. (2021), Singh et al. (2018)
GS-inhibitor (HRAC 10)	GS2 amplification	United States	Carvalho-Moore et al. (2022), Noguera et al. (2022)
PPO-inhibitor (HRAC 14)	R128G/M	United States	Giacomini et al., 2017
	ΔG210	United States	Salas et al., 2016
	V361A	United States	Nie et al. (2023)
	G399A	United States	Rangani et al. (2019)
	Enhanced detoxification	United States	Borgato et al. (2024), Shyam et al. (2021), Varanasi et al. (2018)
VLCA-inhibitor (HRAC 15)	Enhanced detoxification	United States	Brabham et al. (2019), Hwang et al. (2023)
HPPD-inhibitor (HRAC 27)	Enhanced detoxification	United States	Küpper, Peter, et al. (2018), Nakka, Godar, Wani, et al. (2017)

^aLetters indicate amino acids, and numbers indicate position in the target site enzyme. Letters separated by a slash indicate more than one amino acid substitution has been identified. ND indicates the resistance mechanisms have not been elucidated.

^bIn cases where resistance has been identified in a publication, but information on the resistance mechanisms was elucidated in a different manuscript, both references are displayed in the table.

et al., 2014). However, in a cotton field, an experiment suggested that individuals possessing high *EPSPS* gene copy number exhibit reduced fitness (Cahoon et al., 2022). It is important to mention that this reduction was not translated into differences in interference with crop plants. Inheritance studies revealed that *EPSPS*-CNV is nuclear and dominant but does not follow the single gene segregation model (Gaines et al., 2010; Giacomini et al., 2019; Mohseni-Moghadam et al., 2013). Interestingly, fluorescence in situ hybridization (FISH) revealed that *EPSPS* copies are associated with all chromosomes in resistant plants (Gaines et al., 2010). Further studies revealed that about half of the *EPSPS* copies are present in the form of an extra chromosomal circular DNA (eccDNA), physically attached but not inserted into the genome (Koo et al., 2018). The *EPSPS*-eccDNA displays unique structural polymorphisms, with divergence occurring as consequence of duplication and deletion events. Furthermore, the *EPSPS* cassette carries an autonomous replication sequence, allowing it to duplicate independently from cell division (Gaines et al., 2019; Koo et al., 2018; Molin et al., 2020).

Various investigations into the origins and spread of the *EPSPS*-CNV in *A. palmeri* found that the *EPSPS*-eccDNA is conserved among resistant populations from geographically distant regions, suggesting a single evolutionary event that quickly spread to multiple parts of the US (Küpper, Manmathan, et al., 2018; Molin et al., 2017; Molin et al., 2018; Molin et al., 2020). It has also been observed that while the *EPSPS*-eccDNA is conserved between resistant plants, the genetic background is not, suggesting that eccDNA is inherited separately and able to backcross into locally adapted plants (Gaines et al., 2019; Molin et al., 2020). It has been hypothesized that glyphosate-resistant *A. palmeri* seeds are often dispersed via agricultural machinery, or wind, since they can travel long distances (Oliveira et al., 2018; Sarangi et al., 2017; Sosnoskie et al., 2012).

Amaranthus palmeri can also produce seed through apomixis (Ribeiro et al., 2014). Rates of apomixis may be governed by environmental conditions, thus, putative apomictic female plants are not always present in populations of dioecious species (Hojsgaard & Hörandl, 2019). Ribeiro et al. (2014) confirmed that reproductively isolated glyphosate-sensitive and -resistant female *A. palmeri* plants produced viable seeds, and that resistance was inherited in apomictic progenies, suggesting that *A. palmeri* can produce seeds both sexually and asexually (via apomixis), and that apomixis may enhance the stability of the glyphosate resistance trait in resistant populations in the absence of reproductive partners. These results suggested that inheritance of glyphosate resistance was more influenced by the female than the male parent, although the multiple copies of *EPSPS* are amplified in the nuclear genome. However, it is important to mention that the authors were not able to confirm apomictic progeny using DNA markers. In contrast, two independent studies conducted in glyphosate-resistant *A. palmeri* populations (multiple copies of *EPSPS*) by Mohseni-Moghadam et al. (2013) and Chandi et al. (2012) reported no differences in progeny from reciprocal crosses and concluded that *EPSPS* gene copy number is under nuclear control with no maternal effects. Later, the *EPSPS*-eccDNA was characterized, indicating polymorphisms and the presence of an autonomous replication sequence,

and demonstrating the random segregation in somatic and gametic cells, regardless of the sex of progenitors (Koo et al., 2018).

5 | THE ROLE OF *A. PALMERI* BREEDING SYSTEM IN PLANT ADAPTATION

The sexual characteristics of a plant may have a large influence over whether it will be successful when invading new environments. High pollen producing invasive species may accelerate the replacement of monoecious, native species. For *Mercurialis annua*, for example, Buggs and Pannell (2006) reported that dioecious, diploid populations displaced monoecious, hexaploid populations over a period of four decades. Hybridization may also facilitate the introduction of alien species to new habitats and increase their adaptability in the new environment. In some cases, such as the case of *Senecio* species in Australia, the replacement of native by invasive species facilitated by hybridization may cause the extinction of the former (Prentis et al., 2007). Hybrids of the invasive and local species may serve as a bridge for further colonization by the invasive. Mesgaran et al. (2016) proposed that, over time, the colonizer will dominate the area through backcrossing, and the local species is eventually replaced by the colonizer. Hybridization can also introduce adaptive traits that contribute to the expansion of the invasive species into new habitats.

There are many traits associated with agricultural colonization. Phenotypic plasticity is associated with a species' ability to colonize new environment compared to less plastic species or native populations. Phenotypic plasticity is the ability of a plant genotype to express different phenotypes in response to biotic and abiotic stimuli (West-Eberhard, 1989). Plasticity can 'buy time' for a population until novel genetic variation arises at the population level with more adapted variants (Fox et al., 2019). In a meta-analysis with 75 studies (Davidson et al., 2011), it was observed that in nearly all native/non-native comparisons, non-native populations had greater phenotypic plasticity for traits like biomass, nitrogen use efficiency, and water use efficiency. In the case of *A. palmeri*, studies have documented the capacity for the introduction of herbicide resistance genes via hybridization (Section 2.2). However, other adaptive traits such as drought tolerance, high seed production, or low seed dormancy may contribute to the expansion of *A. palmeri* as a colonizing species in new environments. For *A. palmeri*, drought conditions induced flowering asynchrony between male and female plants (Mesgaran et al., 2021). Water stress delayed timing of anthesis in males to a greater extent than females, thus, increasing the anthesis mismatch between the two sexes up to 7 days. Progeny seeds originating from water-stressed plants were heavier and larger than those from well-watered plants. Furthermore, plants exposed to water stress also produced seeds that were approximately 30% less dormant than seeds from plants grown under control conditions (Matzrafi et al., 2020). It was also shown that *A. palmeri* plants experiencing drought during growth produced seeds that can germinate under drier conditions. These traits may suggest that although the number of seeds produced could be lower due to

flowering asynchrony, seeds would be more fit for harsh conditions and thus more invasive and competitive.

Apomixis may also facilitate the colonization of dioecious species in new habitats. Dupont (2002) suggested that apomixis in the dioecious species *Lindera glauca* serves as a strategy to establish a population in the Japanese islands in the absence of male plants. This was also suggested as a mechanism to ensure successful invasion such as in *Chondrilla juncea*, which primarily reproduces clonally via autonomous gametophytic apomixis, and sexual reproduction occur mainly in the native range (Gaskin et al., 2013). Environmental conditions may trigger changes in reproduction systems as well as sex dimorphism in different plant species. Response to stress factors may also alter female: male ratio in dioecious species. Changes in the male: female ratio may be related to plant mortality because of different environmental conditions. In *Humulus scandens* and *Silene otitis*, males grown under drought conditions maintain lower photosynthesis and show greater photosynthetic damage, thus showing higher drought sensitivity and mortality than females (Liu & Duan, 2013; Soldaat et al., 2000). The female skew ratio may be due to apomixis or autopollination, the spontaneous generation of male flowers on otherwise female plants. While hermaphroditism has been observed in the dioecious species *Cannabis sativa* L. (hemp) (Punja & Holmes, 2020), to the best of our knowledge it has not been documented in *A. palmeri* or other dioecious *Amaranthus*. Thus, in cases where no male plants can be found, female plants producing seeds via apomixis would result in a completely female progeny.

6 | EFFECTS OF THE BREEDING SYSTEM ON MANAGEMENT OPTIONS

6.1 | Selection pressure in agricultural systems and its effect on *A. palmeri* distribution

In agricultural fields, the frequency of hybridization events is determined by biological traits, such as species compatibility, seed viability and vigour, performance of progeny, and weed management strategy (i.e., prevention of new introduction and weed control prior to anthesis). How far a pollen grain can travel is dictated by environmental conditions such as relative humidity, wind speed and direction, temperature, physical characteristics of the *A. palmeri* donor and receptor (e.g., pollen size and density, plant size), and field obstacles along the way (e.g., a high versus low stature crop). *Amaranthus palmeri* is compatible with other species in the genus, such as *A. spinosus*, *A. tuberculatus*, and *A. hybridus*, although they display varied rates of hybridization (Federico et al., 2007; Gaines et al., 2012). Many studies have focused on the transfer of herbicide resistance alleles from *A. palmeri* to other closely related species. It has been demonstrated, for example, that copy number variation in *EPSPS* can confer glyphosate resistance in *A. spinosus* as a result of pollen transfer from glyphosate-resistant *A. palmeri* (Nandula et al., 2014).

Interspecific hybridization of *A. palmeri* with other closely related species is well documented. Progeny fertility largely depends on

genomic compatibility. Most species within the *Amaranthus* genus exhibit 32 or 34 somatic chromosomes, and abnormal chromosome pairing can occur regardless of chromosome number. *Amaranthus palmeri* ($2n = 34$) and *A. tuberculatus* ($2n = 32$), for example, can intercross at low frequency, and produce a progeny that is broadly sterile, but not completely (Federico et al., 2007). The authors observed that surviving individuals were triploid and could be backcrossed with the *A. tuberculatus* parent. The mechanism leading to hybrid mortality was suggested to be hemizygosity or to hybrid dysgenesis. Another example is hybridization with *A. hybridus*, but also at low frequency (less than 0.01%) (Gaines et al., 2012). Conversely, crosses with *A. spinosus* (Gaines et al., 2012) and *A. rudis* (Wetzel et al., 1999) can result in fertile hybrids. Molin et al. (2016) found that an ALS-inhibitor resistant *A. spinosus* $\times$ *A. palmeri* population had a mutated ALS from the *A. palmeri* progenitor. Although hybrids have been extensively analysed in the context of herbicide resistance (Molin et al., 2016; Oliveira et al., 2018), much less is known about other fitness traits of the progeny.

While they are all members of the same subgenus *Amaranthus*, species have distinctive traits that associate with their centre of origin (Table 2). *Amaranthus retroflexus* and *A. hybridus*, for example, are weeds of riverbanks and adapted to moister regions (Sauer, 1967). Flowering time is a dominant trait in *A. retroflexus* and could be introgressed into the genetic background of a closely related species (Kulakow & Jain, 1985). As noted by Baker (1974), flowering time and duration are important weediness traits, because it can allow plants to escape human intervention and set seed prior to control events. Germination timing and requirements vary among *Amaranthus* species (Lawrence et al., 2004). Given interspecific hybridization with local *Amaranthus* species is possible, *A. palmeri* populations introduced to Europe are subjected to introgression of new traits that could hinder their management. For example, hybridization with species with a wider range of germination requirements could allow *A. palmeri* to exhibit a non-uniform germination pattern, making its management more difficult in agricultural systems.

Amaranthus palmeri, as previously discussed, is a dioecious species with open pollination by wind, with potential for long-distance pollen dispersal and interspecific hybridization. This mating system, in theory, favours populations to maintain a large genetic diversity because of (i) immense diversity of variants in the gene pool that natural selection can act upon, (ii) potential for evolutionary rescue by hybridization from new introductions (Carlson et al., 2014), and (iii) heterosis. High levels of genome-wide genetic diversity have been observed in a small amount of germplasm with populations from six US states (Küpper, Manmathan, et al., 2018). Although *A. palmeri* is an obligate outcrosser, populations can be maintained even when inbreeding occurs, as observed by Küpper, Manmathan, et al. (2018). Species with similar breeding system such as *A. tuberculatus* also exhibit this feature (Kreiner et al., 2019). Heterosis for enhanced biomass production has been reported in the intercross between *A. cruentus* and *A. hypocondiatus* (Lehmann et al., 1991). Much remains unknown about the early evolutionary mechanisms of adaptation following *A. palmeri* colonization of new environments. However, as discussed

TABLE 2 Scientific name, common name, subgenera, origins, mating system, pollination system, ploidy level and genome size of *Amaranthus* species of agricultural importance as weed species.

Scientific name	Common name	Subgenera	Origin	Mating system	Pollination system	Ploidy level ^a	Genome size ^b (Mbp)
<i>Amaranthus albus</i>	Tumble pigweed	Albersia	North America	Monoecious	Self-pollinated	2n = 2x = 32 or 34	530.3
<i>Amaranthus blitoides</i>	Prostrate pigweed	Albersia	North America	Monoecious	Self-pollinated	2n = 2x = 32	521.9
<i>Amaranthus deflexus</i>	Low amaranth	Albersia	South America	Monoecious	Self-pollinated	2n = 2x = 34	640.2
<i>Amaranthus hybridus</i>	Smooth pigweed	Amaranthus	North, Central, and South America	Monoecious	Self-pollinated	2n = 2x = 32 or 34	503.8
<i>Amaranthus palmeri</i>	Palmer amaranth	Acnida	North America	Dioecious	Cross-pollinated	2n = 2x = 32 or 34	421.8
<i>Amaranthus powellii</i>	Powell's amaranth	Amaranthus	North America	Monoecious	Self-pollinated	2n = 2x = 32 or 34	512.3
<i>Amaranthus retroflexus</i>	Redroot pigweed	Amaranthus	North America	Monoecious	Self-pollinated	2n = 2x = 32 or 34	555.6
<i>Amaranthus spinosus</i>	Spiny amaranth	Acnida	North, Central, and South America	Monoecious	Self-pollinated	2n = 2x = 34	471.6
<i>Amaranthus tuberculatus</i>	Waterhemp	Acnida	North America	Dioecious	Cross-pollinated	2n = 2x = 32	675.6
<i>Amaranthus viridis</i>	Slender amaranth	Albersia	South America	Monoecious	Self-pollinated	2n = 2x = 32 or 34	543.1

^aAssad et al. (2017).

^bStetter and Schmid (2017).

by Küpper, Manmathan, et al. (2018), *A. palmeri* populations from different geographic regions share high genetic similarity, which could be explained, in parts, by strong selection pressure caused by herbicides and, importantly, by intense gene flow between recent introductions and local populations.

6.2 | How can we use characteristics in its breeding system to control *A. palmeri*

The breeding system of *A. palmeri* provide unique management opportunities. Recently, WeedOUT (<https://www.weedout-ibs.com>) has developed a sterile pollen-based weed control strategy for *A. palmeri*. WeedOUT's irradiated pollens compete successfully with naturally occurring *A. palmeri* pollen and fertilizes the weed ovule, which leads to the formation of deformed light weight seeds. Consequently, WeedOUT's novel technology reduces seed germination of treated plants to zero. This technology is based on the previously developed sterile insect technique (SIT), where irradiation is used to sterilize mass-reared insects so that, while remaining sexually competitive, they cannot produce any offspring (Knipling, 1959). It remains to be seen the efficacy of this technique in a practical settings and the evolutionary of resistance/escape by *A. palmeri* populations.

Genetic manipulation of weedy traits is a controversial topic with much interest as it promises to alter adaptive traits at the population level (Barrett et al., 2019). Gene drive, for example, is a genetic manipulation where normal Mendelian inheritance is altered, biasing the segregation to an allele of interest by spreading the fitness-reducing genetic modifications through artificial populations in the wild (Neve, 2018). Although the use of gene drive for weed management is at its infancy, other disciplines have been using this technology commercially. For example, a proprietary self-limiting gene is being used in the Florida Keys, Florida, USA, individuals die before reaching

adulthood when containing the gene drive (Najjar et al., 2017). It should be noted that gene drives are currently being developed for testing in malaria-carrying mosquitoes and for invasive rats in New Zealand (Champer et al., 2021; Gantz et al., 2015). Dioecious *Amaranthus* species are particularly good candidates for gene drives as they rely on sexual reproduction and cross pollination. The National Academy of Sciences of the US suggested *A. palmeri* could be a lead candidate for genetic manipulation using gene drive to fix glyphosate susceptibility alleles in the population (National Academies of Sciences, 2016). Alternatively, a gene drive could be developed to reduce the frequency of female plants in the population, eventually leading to local extinction. Gene drives in natural populations may have unexpected ecological consequences associated with altering populations, because native population may be ecologically important, although mechanisms to limit spillovers have been developed (Barrett et al., 2019; Greenbaum et al., 2021). Since *A. palmeri* is an endemic species in the United States, if gene drives that shift an entire population to extinction are more effective than expected, they may eradicate *A. palmeri* from its native range, disrupting the natural ecosystem and possibly causing irreparable damage. *Amaranthus palmeri* may be the prime example for this technology in plants; however, like all technologies, gene drive may also be thwarted by *A. palmeri*'s rapid adaptation, such as the evolution of apomictic populations (Section 4.2).

Another technology being evaluated for weed control is spray-induced gene silencing also known as 'foliar applied RNA interference (RNAi)'. This non-GMO technology relies on the uptake of small silencing RNAs that interfere with the production of target messenger RNAs (mRNA). RNAi is a natural process that all plants perform for gene expression regulation; however, it can be manipulated in certain ways to lead plants to death. For instance, RNAi molecules could target key mRNAs in the plant metabolism to disrupt metabolic pathways, leading to weed control or turn off herbicide resistance genes, reverting once resistant plants into susceptibility (Mezzetti et al., 2020; Sammons

et al., 2015). Another possibility for dioecious species like *A. palmeri* is to target genes that determine sex, and thereby manipulating plants to all be male, leading to reduced seed production. The major advantage of this technology is the lack of synthetic chemicals and off-target effects as the RNAi technology is specific to the exact allele of the target mRNA. Furthermore, if this selects for polymorphisms in the target mRNA, RNAi molecules can be quickly modified to match the mutation. The major challenge currently being addressed for effective RNAi is how to practically deliver such molecules into plant cells, given plants have multiple protective layers outside of the plasma membrane and small RNAs are typically very charged (Bennett et al., 2020; Dalakouras et al., 2016; Jiang et al., 2014). Some of these limitations, such as rapid degradation by nucleases, could be overcome by use of FANA ASO (2'-Fluoro-arabinonucleic Acid antisense oligonucleotides), as reviewed elsewhere (El-Khoury & Damha, 2021).

7 | FUTURE PERSPECTIVES FOR *A. PALMERI* IN EUROPE

Amaranthus palmeri's range is expected to continue expanding across Europe and the Mediterranean region with the predicted climate change (EPPO, 2020). Several herbicide resistant populations of *A. palmeri* have already been documented in different countries across the region (Matzrafi et al., 2025). Along with the reduced ability to manage *A. palmeri*, due to herbicide resistance evolution, we may expect a reduction in yield production in agricultural crops if *A. palmeri* best management practices are not locally developed, and closely linked with educational resources readily available to producers. Due to the sexual traits of *A. palmeri* listed above, herbicide resistance may spread quickly to susceptible populations as well as stack rapidly in multiple resistant individuals (Gaines et al., 2020). Previous studies have documented the capacity for the introduction of herbicide resistance genes via hybridization in *A. palmeri*. However, other adaptive traits (e.g., drought tolerance, low seed dormancy etc.) may contribute to the expansion of *A. palmeri* as a colonizing species across Europe and the Mediterranean region. Hybrids may play an important role in species introduction and adaptation to new environment. The fact that many other *Amaranthus* species are already highly abundant in agricultural areas in Europe and the Mediterranean region suggests an increased risk for interspecific hybridization and the introgression of new traits could hinder *A. palmeri* management. Several cases of resistance to PSII and ALS inhibitors were documented in populations of *Amaranthus* species in different countries across Europe (Heap, 2023), enhancing the odds for hybridization and rapid resistance evolution.

In order to prevent further invasions and spread of *A. palmeri* to Europe and the Mediterranean region, accurate identification of *A. palmeri* and distinction from closely related *Amaranthus* species is required (Matzrafi et al., 2025). The major driver for weed invasion are import-export trades (Buddenhagen et al., 2021). Thus, apart from weed identification at the whole plant level, seed identification in grain-shipments is highly important to prevent further introduction of

new *A. palmeri* populations. Also, the use of certified seeds could be adopted by farmer groups. Since seeds of different *Amaranthus* species may be very similar in size and shape, different molecular techniques such as ITS2 sequences and restriction enzymes (Wetzel et al., 1999), as well as *A. palmeri*-specific SNPs (Brusa et al., 2021), should be used to distinguish seeds of *A. palmeri* from those of other *Amaranthus* species. Developing these methods as tools for plant protection services and agencies may aid to prevent further introductions of *A. palmeri* to the region through grain-shipments.

Management options for the control of *A. palmeri* may include chemical, mechanical and physical weed management, as well as various agricultural practices such as crop rotation, cover crops, and others (Matzrafi et al., 2025). Due to its superior adaptive abilities, *A. palmeri* may adapt to different management practices and control options. Understanding the breeding system of *A. palmeri* can be greatly beneficial to identify an 'Achilles heel' to further targeting with emerging weed control technologies, such as irradiated pollens, genetic manipulations, and RNAi. However, more studies are needed to assess the efficacy of these technologies under field conditions.

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The authors declare that no new data was collected, presented or analysed for publication of this paper.

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REFERENCES

- Abu-Nasar, J., Wllach, A., Winkler, E., Eizenberg, H. & Matzrafi, M. (2025) Occurrence of *Amaranthus palmeri* in Israeli agriculture. Status of spread and response to glyphosate and tryfloxysulfuron. *Weed Research*, 65, e12616. Available from: <https://doi.org/10.1111/wre.12616>
- Assad, R., Reshi, Z.A., Jan, S. & Rashid, I. (2017) Biology of Amaranths. *The Botanical Review*, 83, 382–436.
- Bagavathiannan, M.V. & Norsworthy, J.K. (2016) Multiple-herbicide resistance is widespread in roadside palmer amaranth populations. *PLoS One*, 11, e0148748.
- Baker, H.G. (1974) The evolution of weeds. *Annual Review of Ecology and Systematics*, 5(1), 1–24.
- Barrett, L.G., Legros, M., Kumaran, N., Glassop, D., Raghu, S. & Gardiner, D.M. (2019) Gene drives in plants: opportunities and challenges for weed control and engineered resilience. *Proceedings of the Royal Society B: Biological Sciences*, 286(1911), 1515.
- Barrett, S.C.H. & Harder, L.D. (2017) The ecology of mating and its evolutionary consequences in seed plants. *Annual Review of Ecology, Evolution, and Systematics*, 48(1), 135–157.
- Basinger, N.T., Jennings, K.M., Monks, D.W., Jordan, D.L., Everman, W.J., Hestir, E.L. et al. (2019) Large crabgrass (*Digitaria sanguinalis*) and Palmer amaranth (*Amaranthus palmeri*) intraspecific and interspecific interference in soybean. *Weed Science*, 67(6), 649–656.
- Bennett, M., Deikman, J., Hendrix, B. & Iandolino, A. (2020) Barriers to efficient foliar uptake of dsRNA and molecular barriers to dsRNA

- activity in plant cells. *Frontiers in Plant Science*, 11, 816. Available from: <https://doi.org/10.3389/fpls.2020.00816>
- Berger, S., Madeira, P., Ferrell, J., Gettys, L., Morichetti, S., Cantero, J. et al. (2016) Palmer Amaranth (*Amaranthus palmeri*) identification and documentation of ALS-resistance in Argentina. *Weed Science*, 64(2), 312–320.
- Bobadilla, L.K., Giacomini, D.A., Hager, A.G. & Tranel, P.J. (2022) Characterization and inheritance of dicamba resistance in a multiple-resistant waterhemp (*Amaranthus tuberculatus*) population from Illinois. *Weed Science*, 70(1), 4–13.
- Borgato, E.A., Thiagarayaselvam, A., Peterson, D.E., Hay, M.M., Dille, J.A. & Jugulam, M. (2024) Metabolic resistance to protoporphyrinogen oxidase-inhibitor herbicides in a Palmer amaranth population from Kansas. *Journal of Agricultural and Food Chemistry*, 72, 5122–5132. Available from: <https://doi.org/10.1021/acs.jafc.3c05333>
- Brabham, C., Norsworthy, J., Houston, M., Varanasi, V. & Barber, T. (2019) Confirmation of S-metolachlor resistance in Palmer amaranth (*Amaranthus palmeri*). *Weed Technology*, 33(5), 720–726.
- Bravo, W., Leon, R.G., Ferrell, J.A., Mulvaney, M.J. & Wood, C.W. (2017) Differentiation of life-history traits among palmer amaranth populations (*Amaranthus palmeri*) and its relation to cropping systems and glyphosate sensitivity. *Weed Science*, 65(3), 339–349.
- Brusa, A., Patterson, E.L., Gaines, T.A., Dorn, K., Westra, P., Sparks, C.D. et al. (2021) A needle in a seedstack: an improved method for detection of rare alleles in bulk seed testing through KASP. *Pest Management Science*, 77(5), 2477–2484.
- Brusamarello, A.P., Oliveira, P.H., Trezzi, M.M., Finatto, T., Pagnoncelli, F.D.B., Jr. & Vidal, R.A. (2020) Inheritance of fomesafen and imazethapyr resistance in a multiple herbicide-resistant *Euphorbia heterophylla* population. *Weed Research*, 60, 278–286.
- Buddenhagen, C.E., Rubenstein, J.M., Hampton, J.G. & Rolston, M.P. (2021) The phytosanitary risks posed by seeds for sowing trade networks. *PLoS One*, 16(11), e0259912.
- Buggs, R.J.A. & Pannell, J.R. (2006) Rapid displacement of a monoecious plant lineage is due to pollen swamping by a dioecious relative. *Current Biology*, 16(10), 996–1000.
- Cahoon, C., Jordan, D., Tranel, P.J., York, A.C., Riggins, C., Seagrover, R. et al. (2022) In-field assessment of EPSPS amplification on fitness cost in mixed glyphosate-resistant and glyphosate-sensitive populations of Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 70(6), 663–668.
- Cao, J., Wu, Q., Wan, F., Guo, J. & Wang, R. (2022) Reliable and rapid identification of glyphosate-resistance in the invasive weed *Amaranthus palmeri* in China. *Pest Management Science*, 78(6), 2173–2182.
- Carlson, S.M., Cunningham, C.J. & Westley, P.A.H. (2014) Evolutionary rescue in a changing world. *Trends in Ecology & Evolution*, 29, 521–530.
- Carvalho-Moore, P., Norsworthy, J.K., González-Torralva, F., Hwang, J.I., Patel, J.D., Barber, T.L. et al. (2022) Unraveling the mechanism of resistance in a glufosinate-resistant Palmer amaranth (*Amaranthus palmeri*) accession. *Weed Science*, 70(4), 370–379.
- Chahal, P.S., Irmak, S., Jugulam, M. & Jhala, A.J. (2018) Evaluating effect of degree of water stress on growth and fecundity of palmer amaranth (*Amaranthus palmeri*) using soil moisture sensors. *Weed Science*, 66(6), 738–745.
- Champer, S.E., Oakes, N., Sharma, R., García-Díaz, P., Champer, J. & Messer, P.M. (2021) Modeling CRISPR gene drives for suppression of invasive rodents using a supervised machine learning framework. *PLoS Computational Biology*, 17(12), e1009660.
- Chandi, A., Milla-Lewis, S.R., Giacomini, D., Westra, P., Preston, C., Jordan, D.L. et al. (2012) Inheritance of evolved glyphosate resistance in a North Carolina palmer amaranth (*Amaranthus palmeri*) biotype. *International Journal of Agronomy*, 12, 1–7. Available from: <https://doi.org/10.1155/2012/176108>
- Culpepper, A., Grey, T., Vencill, W., Kichler, J., Webster, T., Brown, S. et al. (2006) Glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*) confirmed in Georgia. *Weed Science*, 54(4), 620–626.
- Délye, C. (2013) Unravelling the genetic bases of non-target-site-based resistance (NTSR) to herbicides: a major challenge for weed science in the forthcoming decade. *Pest Management Science*, 69(2), 176–187.
- Dalakouras, A., Wassenegger, M., McMillan, J.N., Cardoza, V., Maegele, I., Dadami, E. et al. (2016) Induction of silencing in plants by high-pressure spraying of in vitro-synthesized small RNAs. *Frontiers in Plant Science*, 7, 1327.
- Davidson, A.M., Jennions, M. & Nicotra, A.B. (2011) Do invasive species show higher phenotypic plasticity than native species and, if so, is it adaptive? A meta-analysis. *Ecology Letters*, 14, 419–431.
- Davis, D., Letarte, J., Grainger, C.M., Rajcan, I. & Tardiff, F.J. (2019) Widespread herbicide resistance in pigweed species in Ontario carrot production is due to multiple photosystem II mutations. *Canadian Journal of Plant Science*, 100(1), 56–67.
- Dominguez-Valenzuela, J.A., Gherekhloo, J., Fernández-Moreno, P.T., Cruz-Hipolito, H.E., Alcántara de la Cruz, R., Sánchez-González, E. et al. (2017) First confirmation and characterization of target and non-target site resistance to glyphosate in Palmer amaranth (*Amaranthus palmeri*) from Mexico. *Plant Physiology and Biochemistry*, 115, 212–218.
- Dumont, M., Letarte, J. & Tardif, F.J. (2016) Identification of a psbA mutation (valine₂₁₉ to isoleucine) in Powell amaranth (*Amaranthus powellii*) conferring resistance to linuron. *Weed Science*, 64(1), 6–11.
- Dupont, Y. (2002) Evolution of apomixis as a strategy of colonization in the dioecious species *Lindera glauca* (Lauraceae). *Population Ecology*, 44, 293–297.
- El-Khoury, R. & Damha, M.J. (2021) 2'-Fluoro-arabinonucleic acid (FANA): a versatile tool for probing biomolecular interactions. *Accounts of Chemical Research*, 54(9), 2287–2297.
- EPPO. (2020) *Pest risk analysis for Amaranthus palmeri*. Paris: EPPO. Available from: <https://gd.eppo.int/taxon/AMAPA/documents>
- Federico, T., Danman, Z., Andrew, J.W., Walter, J.R., Tatum, T.C., Rayburn, A.L. et al. (2007) Nonhybrid progeny from crosses of dioecious amaranths: implications for gene-flow research. *Weed Science*, 55(2), 119–122.
- Foes, M.J., Liu, L., Vigue, G., Stoller, E.W., Wax, L.M. & Tranel, P.J. (1999) A kochia (*Kochia scoparia*) biotype resistant to triazine and ALS inhibiting herbicides. *Weed Science*, 47(1), 20–27.
- Foster, D.C. & Steckel, L.E. (2022) Confirmation of dicamba-resistant palmer amaranth in Tennessee. *Weed Technology*, 36(6), 777–780.
- Fox, R.J., Donelson, J.M., Schunter, C., Ravasi, T. & Gaitán-Espitia, J.D. (2019) Beyond buying time: the role of plasticity in phenotypic adaptation to rapid environmental change. *Philosophical Transactions of the Royal Society B*, 374, e20180174.
- Franssen, A., Skinner, D., Al-Khatib, K., Horak, M. & Kulakow, P. (2001) Interspecific hybridization and gene flow of ALS resistance in *Amaranthus* species. *Weed Science*, 49(5), 598–606.
- Gaines, T.A., Duke, S.O., Morran, S., Rigon, C.A.G., Tranel, P.J., Küpper, A. et al. (2020) Mechanisms of evolved herbicide resistance. *The Journal of Biological Chemistry*, 295(30), 10307–10330.
- Gaines, T.A., Patterson, E.L. & Neve, P. (2019) Molecular mechanisms of adaptive evolution revealed by global selection for glyphosate resistance. *The New Phytologist*, 223(4), 1770–1775.
- Gaines, T.A., Shaner, D.L., Ward, S.M., Leach, J.E., Preston, C. & Westra, P. (2011) Mechanism of resistance of evolved glyphosate-resistant palmer amaranth (*Amaranthus palmeri*). *Journal of Agricultural and Food Chemistry*, 59(11), 5886–5889.
- Gaines, T.A., Slavov, G.T., Hughes, D., Küpper, A., Sparks, C.D., Oliva, J. et al. (2021) Investigating the origins and evolution of a glyphosate-resistant weed invasion in South America. *Molecular Ecology*, 30(21), 5360–5372.
- Gaines, T.A., Ward, S.M., Bukun, B., Preston, C., Leach, J.E. & Westra, P. (2012) Interspecific hybridization transfers a previously unknown glyphosate resistance mechanism in *Amaranthus* species. *Evolutionary Applications*, 5(1), 29–38.

- Gaines, T.A., Zhang, W., Wang, D., Bukun, B., Chisholm, S.T., Shaner, D.L. et al. (2010) Gene amplification confers glyphosate resistance in *Amaranthus palmeri*. *Proceedings of the National Academy of Sciences*, 107(3), 1029–1034.
- Gantz, V.M., Jasinskiene, N., Tatarenkova, O., Fazekas, A., Macias, V.M., Bier, E. et al. (2015) Highly efficient Cas9-mediated gene drive for population modification of the malaria vector mosquito *Anopheles stephensi*. *Proceedings of the National Academy of Sciences*, 112(49), E6736–E6743.
- Gaskin, J.F., Schwarzländer, M., Kinter, C.L., Smith, J.F. & Novak, S.J. (2013) Propagule pressure, genetic structure, and geographic origins of *Chondrilla juncea* (Asteraceae): an apomitic invader on three continents. *American Journal of Botany*, 100(9), 1871–1882.
- Giacomini, D., Westra, P. & Ward, S. (2014) Impact of genetic background in fitness cost studies: an example from glyphosate-resistant Palmer amaranth. *Weed Science*, 62(1), 29–37.
- Giacomini, D., Westra, P. & Ward, S. (2019) Variable inheritance of amplified EPSPS gene copies in glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 67(2), 176–182.
- Giacomini, D.A., Umphres, A.M., Nie, H., Mueller, T.C., Steckel, L.E., Young, B.G. et al. (2017) Two new PPX2 mutations associated with resistance to PPO-inhibiting herbicides in *Amaranthus palmeri*. *Pest Management Science*, 73(8), 1559–1563.
- Gonçalves Netto, A., Nicolai, M., Carvalho, S.J.P., Borgato, E.A. & Christoffoleti, P.J. (2016) Multiple resistance of *Amaranthus palmeri* to ALS and EPSPS inhibiting herbicides in the state of Mato Grosso, Brazil. *Planta Daninha*, 34, 581–587.
- Gossett, B.J., Murdock, E.C. & Toler, J.E. (1992) Resistance of palmer amaranth (*Amaranthus palmeri*) to the dinitroaniline herbicides. *Weed Technology*, 6(3), 587–591.
- Greenbaum, G., Feldman, M.W., Rosenberg, N.A. & Kim, J. (2021) Designing gene drives to limit spillover to non-target populations. *PLoS Genetics*, 17, e1009278.
- Hare, A.T., Jordan, D.L., Leon, R.G., Edmisten, K.L., Post, A.R., Mahoney, D.J. et al. (2019) Impact of weed management on peanut yield and weed populations the following year. *Peanut Science*, 46(2), 182–190.
- Hart, S.E., Saunders, J.W. & Penner, D. (1993) Semidominant nature of monogenic sulfonylurea herbicide resistance in sugarbeet (*Beta vulgaris*). *Weed Science*, 41(3), 317–324.
- Heap, I. (2023) The international herbicide-resistant weed database. Available from: www.weedscience.org
- Hirschberg, J. & McIntosh, L. (1983) Molecular basis of herbicide resistance in *Amaranthus hybridus*. *Science*, 222(4630), 1346–1349.
- Højsgaard, D. & Hörandl, E. (2019) The rise of apomixis in natural plant populations. *Frontiers in Plant Science*, 10, 358. Available from: <https://doi.org/10.3389/fpls.2019.00358>
- Horak, M. & Loughin, T. (2000) Growth analysis of four *Amaranthus* species. *Weed Science*, 48(3), 347–355.
- Horak, M.J. & Peterson, D.E. (1995) Biotypes of palmer amaranth (*Amaranthus palmeri*) and common waterhemp (*Amaranthus rudis*) are resistant to imazethapyr and thifensulfuron. *Weed Technology*, 9(1), 192–195.
- Hwang, J.I., Norsworthy, J.K., Carvalho-Moore, P., Barber, L.T., Butts, T.R. & McElroy, J.S. (2023) Exploratory analysis on herbicide metabolism and very-long-chain fatty acid production in metolachlor-resistant Palmer amaranth (*Amaranthus palmeri* S. Wats.). *Journal of Agricultural and Food Chemistry*, 71, 6014–6022. Available from: <https://doi.org/10.1021/acs.jafc.3c00196>
- Jeschke, M.R., Tranel, P.J. & Rayburn, A.L. (2003) DNA content analysis of smooth pigweed (*Amaranthus hybridus*) and tall waterhemp (*A. tuberculatus*): implications for hybrid detection. *Weed Science*, 51(1), 1–3.
- Jhala, A.J., Norsworthy, J.K., Ganie, Z.A., Sosnoskie, L.M., Beckie, H.J., Mallory-Smith, C.A. et al. (2021) Pollen-mediated gene flow and transfer of resistance alleles from herbicide-resistant broadleaf weeds. *Weed Technology*, 35(2), 173–187.
- Jiang, L., Ding, L., He, B., Shen, J., Xu, Z., Yin, M. et al. (2014) Systemic gene silencing in plants triggered by fluorescent nanoparticle-delivered double-stranded RNA. *Nanoscale*, 6, 9965–9969.
- Küpfer, A., Borgato, E.A., Patterson, E., Gonçalves Netto, A., Nicolai, M., Carvalho, S.J.P. et al. (2017) Multiple resistance to glyphosate and acetolactate synthase inhibitors in Palmer amaranth (*Amaranthus palmeri*) identified in Brazil. *Weed Science*, 65(3), 317–326.
- Küpfer, A., Manmathan, H.K., Giacomini, D., Patterson, E.L., McCloskey, W.B. & Gaines, T.A. (2018) Population genetic structure in glyphosate-resistant and -susceptible Palmer amaranth (*Amaranthus palmeri*) populations using genotyping-by-sequencing (GBS). *Frontiers in Plant Science*, 9, 29. Available from: <https://doi.org/10.3389/fpls.2018.00029>
- Küpfer, A., Peter, F., Zöllner, P., Lorentz, L., Tranel, P.J., Beffa, R. et al. (2018) Tembotrione detoxification in 4-hydroxyphenylpyruvate dioxygenase (HPPD) inhibitor-resistant Palmer amaranth (*Amaranthus palmeri* S. Wats.). *Pest Management Science*, 74(10), 2325–2334.
- Kanatas, P., Tataridas, A., Dellaportas, V. & Travlos, I. (2021) First report of *Amaranthus palmeri* S. Wats. in cotton, maize and sorghum in Greece and problems with its management. *Agronomy*, 11(9), 1721.
- Kaundun, S.S., Jackson, L.V., Hutchings, S.J., Galloway, J., Marchegiani, E., Howell, A. et al. (2019) Evolution of target-site resistance to glyphosate in an *Amaranthus palmeri* population from Argentina and its expression at different plant growth temperatures. *Plants*, 8(11), 512.
- Knipling, E.F. (1959) Sterile-male method of population control. *Science*, 130, 902–904.
- Kohrt, J.R., Sprague, C.L., Nadakuduti, S.S. & Douches, D. (2017) Confirmation of a three-way (glyphosate, ALS, and atrazine) herbicide-resistant population of Palmer amaranth in Michigan. *Weed Science*, 65, 327–338.
- Koo, D.H., Molin, W.T., Saski, C.A., Jiang, J., Putta, K., Jugulam, M. et al. (2018) Extrachromosomal circular DNA-based amplification and transmission of herbicide resistance in crop weed *Amaranthus palmeri*. *Proceedings of the National Academy of Sciences*, 115(13), 3332–3337.
- Kreiner, J.M., Giacomini, D.A., Bemm, F., Waithaka, B., Regalado, J., Lanz, C. et al. (2019) Multiple modes of convergent adaptation in the spread of glyphosate-resistant *Amaranthus tuberculatus*. *Proceedings of the National Academy of Sciences*, 116, 21076–21084.
- Kulakow, P.A. & Jain, S.K. (1985) The inheritance of flowering time in *Amaranthus* species. *Journal of Genetics*, 64, 85–100.
- Larran, A.S., Palmieri, V.E., Perotti, V.E., Lieber, L., Tuesca, D. & Permingeat, H.R. (2017) Target-site resistance to acetolactate synthase (ALS)-inhibiting herbicides in *Amaranthus palmeri* from Argentina. *Pest Management Science*, 73(12), 2578–2584.
- Lawrence, E.S., Christy, L.S., Edward, W.S. & Loyd, M.W. (2004) Temperature effects on germination of nine *Amaranthus* species. *Weed Science*, 52(2), 217–221.
- Lehmann, J.W., Clark, R.L. & Frey, K.J. (1991) Biomass heterosis and combining ability in interspecific and intraspecific matings of grain amaranths. *Crop Science*, 31, 1111–1116.
- Liu, J. & Duan, J. (2013) *Humulus scandens* gender differences in response to water stress in the vegetative growth stage. *Acta Prataculturae Sinica*, 22, 243–249.
- Liu, M., Korpeläinen, H. & Li, C. (2021) Sexual differences and sex ratios of dioecious plants under stressful environments. *Journal of Plant Ecology*, 14(5), 920–933.
- Manicardi, A., Scarabel, L., Llenes, J.M., Montull, J.M., Osuna, M.D., Torra Farré, J. et al. (2025) Genetic basis and origin of resistance to acetolactate synthase inhibitors in *Amaranthus palmeri* from Spain and Italy. *Pest Management Science*, 79(12), 4886–4896.
- Manicardi, A., Milani, A., Scarabel, L., Mora, G., Recasens, J., Llenes, J.M. et al. (2025) First report of glyphosate resistance in an *Amaranthus palmeri* population from Europe. *Weed Research*, 65, e12579. Available from: <https://doi.org/10.1111/wre.12579>
- Massinga, R.A., Currie, R.S., Horak, M.J. & Boyer, J. (2001) Interference of Palmer amaranth in corn. *Weed Science*, 49(2), 202–208.

- Matzrafi, M., Osipitan, O.A., Ohadi, S. & Mesgaran, M.B. (2020) Under pressure: maternal effects promote drought tolerance in progeny seed of Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 69(1), 31–38.
- Matzrafi, M., Scarabel, L., Milani, A., Iamónico, D., Torra, J., Recasens, J. et al. (2025) *Amaranthus palmeri* S. Watson: a new threat to agriculture in Europe and the Mediterranean region. *Weed Research*, 65, e12596. Available from: <https://doi.org/10.1111/wre.12596>
- Mennan, H., Kaya-Altóp, E., Belvaux, X., Brants, I., Zandstra, B.H., Jabran, K. et al. (2021) Investigating glyphosate resistance in *Amaranthus palmeri* biotypes from Turkey. *Phytoparasitica*, 49, 1043–1052.
- Mesgaran, M.B., Lewis, M.A., Ades, P.K., Donohue, K., Ohadi, S., Li, C. et al. (2016) Hybridization can facilitate species invasions, even without enhancing local adaptation. *Proceedings of the National Academy of Sciences*, 113(36), 10210–10214.
- Mesgaran, M.B., Matzrafi, M. & Ohadi, S. (2021) Sex dimorphism in dioecious Palmer amaranth (*Amaranthus palmeri*) in response to water stress. *Planta*, 254(1), 17.
- Mezzetti, B., Smagghe, G., Arpaia, S., Christiaens, O., Dietz-Pfeilstetter, A., Jones, H. et al. (2020) RNAi: what is its position in agriculture? *Journal of Pest Science*, 93, 1125–1130.
- Milani, A., Panozzo, S., Farinati, S., Iamónico, D., Sattin, M., Loddo, D. et al. (2021) Recent discovery of *Amaranthus palmeri* S. Watson in Italy: characterization of ALS-resistant populations and sensitivity to alternative herbicides. *Sustainability*, 13(13), 7003.
- Ming, R., Wang, J., Moore, P.H. & Paterson, A.H. (2007) Sex chromosomes in flowering plants. *American Journal of Botany*, 94(2), 141–150.
- Mohseni-Moghadam, M., Schroeder, J. & Ashigh, J. (2013) Mechanism of resistance and inheritance in glyphosate resistant Palmer amaranth (*Amaranthus palmeri*) populations from New Mexico, USA. *Weed Science*, 61(4), 517–525.
- Molin, W., Nandula, V., Wright, A. & Bond, J. (2016) Transfer and expression of ALS inhibitor resistance from Palmer amaranth (*Amaranthus palmeri*) to an *A. spinosus* × *A. palmeri* hybrid. *Weed Science*, 64(2), 240–247.
- Molin, W.T., Wright, A.A., Lawton-Rauh, A. & Saski, C.A. (2017) The unique genomic landscape surrounding the EPSPS gene in glyphosate resistant *Amaranthus palmeri*: a repetitive path to resistance. *BMC Genomics*, 18(1), 91.
- Molin, W.T., Wright, A.A., VanGessel, M.J., McCloskey, W.B., Jugulam, M. & Hoagland, R.E. (2018) Survey of the genomic landscape surrounding the 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) gene in glyphosate-resistant *Amaranthus palmeri* from geographically distant populations in the USA. *Pest Management Science*, 74(5), 1109–1117.
- Molin, W.T., Yaguchi, A., Blenner, M. & Saski, C.A. (2020) Autonomous replication sequences from the *Amaranthus palmeri* eccDNA replicon enable replication in yeast. *BMC Research Notes*, 13, 1–6.
- Montgomery, J., Giacomini, D. & Tranel, P. (2021) Molecular confirmation of resistance to PPO inhibitors in *Amaranthus tuberculatus* and *Amaranthus palmeri*, and isolation of the G399A PPO2 substitution in *A. palmeri*. *Weed Technology*, 35(1), 99–105.
- Montgomery, J., Sadeque, A., Giacomini, D., Brown, P. & Tranel, P. (2019) Sex-specific markers for waterhemp (*Amaranthus tuberculatus*) and Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 67(4), 412–418.
- Montgomery, J.S., Giacomini, D.A., Weigel, D. & Tranel, P.J. (2021) Male-specific Y-chromosomal regions in waterhemp (*Amaranthus tuberculatus*) and Palmer amaranth (*Amaranthus palmeri*). *The New Phytologist*, 229(6), 3522–3533.
- Moore, J.W., Murray, D.S. & Westerman, R.B. (2004) Palmer amaranth (*Amaranthus palmeri*) effects on the harvest and yield of grain sorghum (*Sorghum bicolor*). *Weed Technology*, 18(1), 23–29.
- Najjar, D.A., Normandin, A.M., Strait, E.A. & Esvelt, K.M. (2017) Driving towards ecotechnologies. *Pathogens and Global Health*, 111(8), 448–458.
- Nakka, S., Godar, A.S., Thompson, C.R., Peterson, D.E. & Jugulam, M. (2017) Rapid detoxification via glutathione S-transferase (GST) conjugation confers a high level of atrazine resistance in Palmer amaranth (*Amaranthus palmeri*). *Pest Management Science*, 73(11), 2236–2243.
- Nakka, S., Godar, A.S., Wani, P.S., Thompson, C.R., Peterson, D.E., Roelofs, J. et al. (2017) Physiological and molecular characterization of hydroxyphenylpyruvate dioxygenase (HPPD)-inhibitor resistance in Palmer amaranth (*Amaranthus palmeri* S.Wats.). *Frontiers in Plant Science*, 8, 555.
- Nakka, S., Thompson, C., Peterson, D. & Jugulam, M. (2017) Target site-based and non-target site-based resistance to ALS inhibitors in Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 65(6), 681–689.
- Nandula, V.K., Wright, A.A., Bond, J.A., Ray, J.D., Eubank, T.W. & Molin, W.T. (2014) EPSPS amplification in glyphosate-resistant spiny amaranth (*Amaranthus spinosus*): a case of gene transfer via interspecific hybridization from glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*). *Pest Management Science*, 70, 1902–1909.
- National Academies of Sciences, Engineering, and Medicine. (2016) *Gene drives on the horizon: advancing science, navigating uncertainty, and aligning research with public values*. Washington, DC: The National Academies Press.
- Neve, P. (2018) Gene drive systems: do they have a place in agricultural weed management? *Pest Management Science*, 74(12), 2671–2679.
- Neves, C.J., Matzrafi, M., Thiele, M., Lorant, A., Mesgaran, M.B. & Stetter, M.G. (2020) Male linked genomic region determines sex in dioecious *Amaranthus palmeri*. *The Journal of Heredity*, 111(7), 606–612.
- Nie, H., Harre, N.T. & Young, B.G. (2023) A new V361A mutation in *Amaranthus palmeri* PPX2 associated with PPO-inhibiting herbicide resistance. *Plants*, 12(9), 1886.
- Noguera, M.M., Porri, A., Werle, I.S., Heiser, J., Brändle, F., Lerchl, J. et al. (2022) Involvement of glutamine synthetase 2 (GS2) amplification and overexpression in *Amaranthus palmeri* resistance to glufosinate. *Planta*, 256(3), 57.
- Norsworthy, J.K., Griffith, G., Griffin, T., Bagavathiannan, M. & Gbur, E.E. (2014) In-field movement of glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*) and its impact on cotton lint yield: evidence supporting a zero-threshold strategy. *Weed Science*, 62(2), 237–249.
- Oliveira, M.C., Gaines, T.A., Patterson, E.L., Jhala, A.J., Irmak, S., Amundsen, K. et al. (2018) Interspecific and intraspecific transference of metabolism-based mesotrione resistance in dioecious weedy *Amaranthus*. *The Plant Journal*, 96(5), 1051–1063.
- Palmieri, V.E., Alvarez, C.E., Permingeat, H.R. & Perotti, V.E. (2022) A122S, A205V, D376E, W574L and S653N substitutions in acetolactate synthase (ALS) from *Amaranthus palmeri* show different functional impacts on herbicide resistance. *Pest Management Science*, 78(2), 749–757.
- Park, K.W. & Mallory-Smith, C.A. (2006) psbA mutation in *Amaranthus retroflexus* confers resistance to several PSII-inhibiting herbicides. *Pest Management Science*, 62, 880–885.
- Patzoldt, W.L., Hager, A.G., McCormick, J.S. & Tranel, P.J. (2006) A codon deletion confers resistance to herbicides inhibiting protoporphyrinogen oxidase. *Proceedings of the National Academy of Sciences*, 103(33), 12329–12334.
- Prentis, P.J., White, E.M., Radford, I.J., Lowe, A.J. & Clarke, A.R. (2007) Can hybridization cause local extinction: a case for demographic swamping of the Australian native *Senecio pinnatifolius* by the invasive *Senecio madagascariensis*? *The New Phytologist*, 176(4), 902–912.
- Priess, G., Norsworthy, J., Godara, N., Mauromoustakos, A., Butts, T., Roberts, T. et al. (2022) Confirmation of glufosinate-resistant palmer amaranth and response to other herbicides. *Weed Technology*, 36(3), 368–372.
- Punja, Z.K. & Holmes, J.E. (2020) Hermaphroditism in marijuana (*Cannabis sativa* L.) inflorescences—impact on floral morphology, seed formation,

- progeny sex ratios, and genetic variation. *Frontiers in Plant Science*, 11, 718. Available from: <https://doi.org/10.3389/fpls.2020.00718>
- Rangani, G., Salas-Perez, R.A., Aponte, R.A., Knapp, M., Craig, I.R., Mietzner, T. et al. (2019) Novel single-site mutation in the catalytic domain of protoporphyrinogen oxidase IX (PPO) confers resistance to PPO-inhibiting herbicides. *Frontiers in Plant Science*, 15, 568.
- Rayburn, A.L., McCloskey, C., Tatum, T.C., Bollero, G.A., Jeschke, M.R. & Tranel, P.J. (2005) Genome size analysis of weedy *Amaranthus* species. *Crop Science*, 45, 2557–2562.
- Reinhardt, C., Vorster, J., Küpper, A., Peter, F., Simelane, A., Friis, S. et al. (2022) A nonnative Palmer amaranth (*Amaranthus palmeri*) population in the Republic of South Africa is resistant to herbicides with different sites of action. *Weed Science*, 70(2), 183–197.
- Ribeiro, D.N., Pan, Z., Duke, S.O., Nandula, V.K., Baldwin, B.S., Shaw, D.R. et al. (2014) Involvement of facultative apomixis in inheritance of EPSPS gene amplification in glyphosate-resistant *Amaranthus palmeri*. *Planta*, 239(1), 199–212.
- Richards, A.J. (1997) *Plant breeding systems*, 2nd edition. London: Chapman & Hall.
- Salas, R.A., Burgos, N.R., Tranel, P.J., Singh, S., Glasgow, L., Scott, R.C. et al. (2016) Resistance to PPO-inhibiting herbicide in palmer amaranth from Arkansas. *Pest Management Science*, 72(5), 864–869.
- Sammons, R.D., Ivashuta, S., Liu, H., Wang, D., Feng, P.C.C., Kouranov, A.Y. et al. (2015) Methods for controlling herbicide-resistant plants (Patent No. US 9,121,022B2).
- Sarangi, D., Tyre, A., Patterson, E.L., Gaines, T.A., Irmak, S., Knezevic, S.Z. et al. (2017) Pollen-mediated gene flow from glyphosate-resistant common waterhemp (*Amaranthus rudis* Sauer): consequences for the dispersal of resistance genes. *Scientific Reports*, 7, 44913.
- Sauer, J.D. (1967) The grain amaranths and their relatives: a revised taxonomic and geographic survey. *Annals of the Missouri Botanical Garden*, 54(2), 103–137.
- Sauer, J.D. (1955) Revision of the dioecious amaranths. *Madrono*, 13, 5–46.
- Sebastian, S.A., Fader, G.M., Ulrich, J.F., Forney, D.R. & Chaleff, R.S. (1989) Semidominant soybean mutation for resistance to sulfonylurea herbicides. *Crop Science*, 29, 1403–1408.
- Sellers, B., Smeda, R., Johnson, W., Kendig, J. & Ellersieck, M. (2003) Comparative growth of six *Amaranthus* species in Missouri. *Weed Science*, 51(3), 329–333.
- Shimono, A., Kanbe, H., Nakamura, S., Ueno, S., Yamashita, J. & Asai, M. (2020) Initial invasion of glyphosate-resistant *Amaranthus palmeri* around grain-import ports in Japan. *Plants People Planet*, 2(6), 640–648.
- Shyam, C., Nakka, S., Putta, K., Cuvaca, I., Currie, R.S. & Jugulam, M. (2021) Genetic basis of chlorsulfuron, atrazine, and mesotrione resistance in a palmer amaranth. *ACS Agricultural Science & Technology*, 1(2), 109–114.
- Shyam, C., Peterson, D.E. & Jugulam, M. (2022) Resistance to 2,4-D in Palmer amaranth (*Amaranthus palmeri*) from Kansas is mediated by enhanced metabolism. *Weed Science*, 70(4), 390–400.
- Singh, S., Singh, V., LawtonRauh, A., Bagavathiannan, M.V. & Roma-Burgos, N. (2018) EPSPS gene amplification primarily confers glyphosate resistance among Arkansas Palmer amaranth (*Amaranthus palmeri*) populations. *Weed Science*, 66(3), 293–300.
- Singh, S., Singh, V., Salas-Perez, R.A., Bagavathiannan, M.V., Lawton-Rauh, A. & Roma-Burgos, N. (2019) Target-site mutation accumulation among ALS inhibitor-resistant Palmer amaranth. *Pest Management Science*, 75(4), 1131–1139.
- Soldaat, L.L., Lorenz, H. & Trefflich, A. (2000) The effect of drought stress on the sex ratio variation of *Silene otites*. *Folia Geobotanica*, 35, 203–210.
- Sosnoskie, L., Webster, T., Kichler, J., MacRae, A., Grey, T. & Culpepper, A. (2012) Pollen-mediated dispersal of glyphosate-resistance in Palmer amaranth under field conditions. *Weed Science*, 60(3), 366–373.
- Stetter, M.G. & Schmid, K.J. (2017) Analysis of phylogenetic relationships and genome size evolution of the *Amaranthus* genus using GBS indicates the ancestors of an ancient crop. *Molecular Phylogenetics and Evolution*, 109, 80–92.
- Torra, J., Royo-Esnal, A., Romano, Y., Osuna, M.D., León, R.G. & Recasens, J. (2020) *Amaranthus palmeri* a new invasive weed in Spain with herbicide resistant biotypes. *Agronomy*, 10, 993.
- Tranel, P. & Wright, T. (2002) Resistance of weeds to ALS-inhibiting herbicides: what have we learned? *Weed Science*, 50(6), 700–712.
- van Wychen, L. (2017) Survey of the most common and troublesome weeds in grass crops, pasture and turf in the United States and Canada. *Weed Science Society of America*, 2017 <http://wssa.net/wp-content/uploads/2017-Weed-Survey>
- Varanasi, V., Brabham, C. & Norsworthy, J. (2018) Confirmation and characterization of non-target site resistance to fomesafen in Palmer amaranth (*Amaranthus palmeri*). *Weed Science*, 66(6), 702–709.
- Vila-Aiub, M.M., Goh, S.S., Gaines, T.A., Han, H., Busi, R., Yu, Q. et al. (2014) No fitness cost of glyphosate resistance endowed by massive EPSPS gene amplification in *Amaranthus palmeri*. *Planta*, 239(4), 793–801.
- Wang, J.L., Klessig, D.F. & Berry, J.O. (1992) Regulation of C4 gene expression in developing amaranth leaves. *Plant Cell*, 4, 173–184.
- Ward, S.M., Webster, T.M. & Steckel, L.E. (2013) Palmer amaranth (*Amaranthus palmeri*): a review. *Weed Technology*, 27(1), 12–27.
- Webster, T.M. & Grey, T.L. (2015) Glyphosate-resistant palmer amaranth (*Amaranthus palmeri*) morphology, growth, and seed production in Georgia. *Weed Science*, 63(1), 264–272.
- West-Eberhard, M.J. (1989) Phenotypic plasticity and the origins of diversity. *Annual Review of Ecology, Evolution, and Systematics*, 20, 249–278.
- Wetzel, D., Horak, M., Skinner, D. & Kulakow, P. (1999) Transfer of herbicide resistance traits from *Amaranthus palmeri* to *Amaranthus rudis*. *Weed Science*, 47(5), 538–543.
- Wright, T.R. & Penner, D. (1998) Corn (*Zea mays*) acetolactate synthase sensitivity to four classes of ALS-inhibiting herbicides. *Weed Science*, 46(1), 8–12.
- Yu, Q. & Powles, S.B. (2010) Evolution in action: plants resistant to herbicides. *Annual Review of Plant Biology*, 61(1), 317–347.
- Zhou, H.P. & Chen, J. (2010) Spatial genetic structure in an understorey dioecious fig species: the roles of seed rain, seed and pollen-mediated gene flow, and local selection. *Journal of Ecology*, 98, 1168–1177.

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